

Title: Console-Based File Explorer in C++ (Linux OS)

Code:

```
#include <iostream>

#include <dirent.h>

#include <sys/stat.h>

#include <unistd.h>

#include <cstring>

#include <fstream>

#include <sstream>

#include <vector>

using namespace std;

void listFiles(const string &path) {

    DIR *dir;

    struct dirent *entry;

    struct stat info;

    dir = opendir(path.c_str());

    if (!dir) {

        cout << " ❌ Cannot open directory: " << path << endl;

        return;

    }

    cout << "\n 📁 Contents of: " << path << "\n";

    cout << "-----\n";

    while ((entry = readdir(dir)) != NULL) {

        string fullPath = path + "/" + entry->d_name;

        stat(fullPath.c_str(), &info);

    }

}
```

```

    if (S_ISDIR(info.st_mode))
        cout << "\033[1;34m[DIR]\033[0m " << entry->d_name << endl; // Blue for folders
    else
        cout << "    " << entry->d_name << endl;
}
closedir(dir);
}

```

```

void changeDirectory(string &currentPath, const string &folder) {
    string newPath = currentPath + "/" + folder;
    if (chdir(newPath.c_str()) == 0) {
        char buffer[1024];
        getcwd(buffer, sizeof(buffer));
        currentPath = buffer;
    } else {
        cout << "❌ Folder not found.\n";
    }
}

```

```

void copyFile(const string &src, const string &dest) {
    ifstream in(src, ios::binary);
    ofstream out(dest, ios::binary);
    if (!in || !out) {
        cout << "❌ Error copying file.\n";
        return;
    }
    out << in.rdbuf();
    cout << "✅ File copied successfully.\n";
}

```

```
}
```

```
void deleteFile(const string &file) {  
    if (remove(file.c_str()) == 0)  
        cout << " 🗑 File deleted.\n";  
    else  
        cout << " ❌ Error deleting file.\n";  
}
```

```
void searchFile(const string &path, const string &filename) {  
    DIR *dir;  
    struct dirent *entry;  
    struct stat info;  
  
    if (!(dir = opendir(path.c_str())))  
        return;  
  
    while ((entry = readdir(dir)) != NULL) {  
        string name = entry->d_name;  
        if (name == "." || name == "..")  
            continue;  
  
        string fullPath = path + "/" + name;  
        stat(fullPath.c_str(), &info);  
  
        if (name.find(filename) != string::npos)  
            cout << " 🔍 Found: " << fullPath << endl;  
  
        if (S_ISDIR(info.st_mode))
```

```

        searchFile(fullPath, filename);
    }
    closedir(dir);
}

void showHelp() {
    cout << "\n  Available Commands:\n";
    cout << "-----\n";
    cout << "ls          - List files in current directory\n";
    cout << "cd <folder>      - Change directory\n";
    cout << "copy <src> <dest> - Copy a file\n";
    cout << "del <file>       - Delete a file\n";
    cout << "search <filename> - Search for a file\n";
    cout << "help           - Show this help message\n";
    cout << "exit          - Exit program\n";
}

```

```

int main() {
    string currentPath;
    char buffer[1024];
    getcwd(buffer, sizeof(buffer));
    currentPath = buffer;

    cout << "  Simple File Explorer (C++)\n";
    showHelp();

    string cmd;
    while (true) {
        cout << "\n[" << currentPath << "]$ ";

```

```

getline(cin, cmd);

if (cmd == "exit") break;
else if (cmd == "ls") listFiles(currentPath);
else if (cmd.rfind("cd ", 0) == 0) changeDirectory(currentPath, cmd.substr(3));
else if (cmd.rfind("copy ", 0) == 0) {
    string src, dest;
    stringstream ss(cmd.substr(5));
    ss >> src >> dest;
    copyFile(src, dest);
}
else if (cmd.rfind("del ", 0) == 0) deleteFile(cmd.substr(4));
else if (cmd.rfind("search ", 0) == 0) searchFile(currentPath, cmd.substr(7));
else if (cmd == "help") showHelp();
else cout << " ? Unknown command. Type 'help' for options.\n";
}

cout << " 🖐️ Exiting File Explorer.\n";
return 0;
}

```

Output:

```

PS C:\Users\rojal\OneDrive\Desktop\FileExplorer> g++ main.cpp -o myexplorer.exe
PS C:\Users\rojal\OneDrive\Desktop\FileExplorer> .\myexplorer.exe
≡f°; Simple File Explorer (C++)

≡fåÿ Available Commands:
-----
ls                - List files in current directory
cd <folder>       - Change directory
copy <src> <dest> - Copy a file
del <file>        - Delete a file
search <filename> - Search for a file
help             - Show this help message
exit            - Exit program

```

```
[C:\Users\rojal\OneDrive\Desktop\FileExplorer]$ ls
Contents of: C:\Users\rojal\OneDrive\Desktop\FileExplorer
-----
[DIR] .
[DIR] ..
explorer.exe
main.cpp
main_backup.cpp
myexplorer.exe
```

```
[C:\Users\rojal\OneDrive\Desktop\FileExplorer]$ copy main.cpp main_backup.c
pp
File copied successfully.
```

```
[C:\Users\rojal\OneDrive\Desktop\FileExplorer]$ del main_backup.cpp
File deleted.
```

```
[C:\Users\rojal\OneDrive\Desktop\FileExplorer]$ search main
Found: C:\Users\rojal\OneDrive\Desktop\FileExplorer/main.cpp
```

```
[C:\Users\rojal\OneDrive\Desktop\FileExplorer]$ help
Available Commands:
-----
ls                - List files in current directory
cd <folder>       - Change directory
copy <src> <dest> - Copy a file
del <file>        - Delete a file
search <filename> - Search for a file
help             - Show this help message
exit            - Exit program
```

```
[C:\Users\rojal\OneDrive\Desktop\FileExplorer]$ exit
Exiting File Explorer.
PS C:\Users\rojal\OneDrive\Desktop\FileExplorer> |
```